Supplementary Material for cfRNA-BrainTrace

Supplementary Methods

S1 Formal validation route

All validation analyses followed the same three-tier architecture used by the software. A query profile was first represented in logCPM- or logTPM-compatible expression space and then projected into Bo2023-like VSD space for Network scoring only. The top three Networks formed the candidate beam. Resolution-group and exact-region candidates were then reranked within that beam using local logCPM-compatible expression. Fold-local Network model construction differed between the LOSO and LOMO validation designs as documented in S7, but projected VSD remained restricted to Network beam generation in both. This design separates broad candidate generation from fine regional interpretation.

S2 Internal validation design

Internal validation used the Bo2023 macaque brain reference. Two settings were evaluated. Leave-one-sample-out validation tested whether the route could recover the held-out sample label when other samples from the reference remained available. Leave-one-monkey-out validation held out all samples from one animal at a time and therefore tested donor-level generalization. Network metrics included all 819 samples because every held-out sample retained a supported Network label. Resolution-group and exact-region metrics required the truth region to remain represented in the corresponding training fold. Five LOSO samples and seven LOMO samples did not meet that region-reference requirement and were excluded only from region-level denominators. Top1, Top3 and median true-rank were used to distinguish exact calls from candidate-list recovery.

S3 External validation design

External validation was limited by the label resolution available in each dataset. AHBA human brain RNA-seq was used for mapped-label validation because its anatomical labels could be harmonized to Network, resolution-group and a subset of exact-region labels. TCGA/BraTS glioma tissue RNA-seq with MRI-derived labels was used only for coarse anatomical consistency because its truth labels are human imaging labels, not Bo2023 macaque exact-region identifiers. GSE189919 was used to test whether an external matrix could be projected into the model gene space; it was not used for accuracy estimation because patient-level anatomical truth was unavailable.

S4 Label harmonization and allowed conclusions

Label harmonization was performed only to the level supported by each dataset. AHBA anatomical labels were mapped to the macaque-derived Network, resolution-group and exact-region hierarchy where a supported mapping existed, and results are interpreted as mapped-label transfer rather than direct anatomical equivalence. TCGA/BraTS labels were derived from human MRI/tumour context and therefore support only coarse tumour-tissue anatomical consistency. Biofluid datasets without patient-level anatomical truth were not used for localization accuracy and are reported only as projection-feasibility or transfer stress tests.

S5 Model artifacts and projected-VSD projector

The repository contains lightweight model artifacts in data/models/, including Network model files, region-resolution dictionaries, route metadata and the reference-fitted projector bo2023_reference_projector_linear_full.npz. The projector stores gene-wise slope, intercept and clipping parameters fitted from reference data. At inference time, a single query profile can be mapped into Bo2023-like projected-VSD space using these fixed parameters without target-cohort labels or target-cohort distribution fitting. Projected VSD is used only for Network Top3 beam generation; resolution-group and exploratory exact-region reranking use logCPM-compatible local expression within that beam.

S6 Example input/output

Synthetic public example files are provided in submission_ready_assets/example_io/ as part of the v0.1.6 public submission release. They document accepted input columns, Network ranked candidates, three-tier JSON output, resolution-group ranked candidates, exact-region ranked candidates and the local generation script. The example files are format and reproducibility aids only; they are not biological validation samples.

S6a Software availability

The manuscript-associated software release is v0.1.6. The public GitHub repository is available at https://github.com/wz7717/cfrna-brain-tracing, and the archived release is cited in the manuscript using the Zenodo version DOI https://doi.org/10.5281/zenodo.20780280; the project concept DOI is https://doi.org/10.5281/zenodo.20773674.

S7 Model development and locked evaluation timeline

The three-tier architecture was fixed before generating the final submission validation tables. External AHBA, TCGA/BraTS and biofluid analyses were not used to select the final route; they were used only for mapped-label transfer evaluation, coarse tumour-tissue consistency assessment and biofluid transfer stress testing. The LOSO implementation used locked Network genes and correlation-ranked projected-VSD Network Top3 candidates. The formal LOMO implementation rebuilt discriminative Network genes and a fold-local pairwise Top1 rescue within each training fold; this rescue could reorder Top1 but did not change the original Network Top3 candidate set. Both implementations used projected VSD only for the Network beam and logCPM-compatible local evidence for downstream reranking. The independent Network-only LOSO and LOMO analyses are reported separately as route-selection evidence.

S8 Figures and tables

Figure source data and synthetic example input/output files are archived in the GitHub/Zenodo release rather than treated as separate journal supplementary files. Supplementary Tables S1-S6 are embedded below so that the formal supplementary PDF can be submitted as a single file containing Supplementary Methods, Supplementary Results and Tables S1-S6. Together, these materials document both how the route was tested and what result each validation setting supports.

Supplementary Results

S1 Internal route selection

The first internal analysis evaluated Network-level candidate generation. Projected VSD achieved Network Top1/Top3 of 58.00%/91.58% in LOSO and 53.72%/91.33% in LOMO, exceeding logCPM baseline and native VSD at Network Top3. Direct exact-region scoring was lower and less stable, especially in LOMO. These results motivated the use of projected VSD for broad Network beam generation and logCPM-compatible expression for downstream local reranking.

S2 Formal internal three-tier validation

In the complete LOSO validation, the formal route achieved Network Top1/Top3 of 58.24%/92.19% across all 819 samples. Resolution-group and exact-region Top3 were 72.36% and 45.33% among 814 reference-supported samples. In complete LOMO validation, Network Top3 was 91.21% across all 819 samples, while resolution-group and exact-region Top3 were 69.09% and 42.36% among 812 reference-supported samples. The earlier LOSO Network value of 92.38% was conditional on the 814 region-evaluable samples and is retained only as a legacy denominator inconsistency, not as the submission result. Median true-rank increased from Network to exact-region levels, consistent with decreasing anatomical certainty at finer resolution. Resolution group is therefore the preferred region-level endpoint, while exact-region output is retained as a candidate ranking.

S3 External validation results

In AHBA, the hybrid route achieved Network Top1/Top3 of 74.68%/94.42%, resolution-group Top1/Top3 of 36.26%/67.03% and exact-region Top1/Top3 of 24.18%/42.86%. Hybrid exact Top3 exceeded logCPM baseline (30.77%) and projected-VSD-only scoring (29.67%). In TCGA/BraTS, hybrid Network Top3 was 40.00% and broad-anatomy Top3 was 64.62%, supporting only coarse anatomical consistency. GSE189919 overlapped 15,622/21,668 projector genes, corresponding to 72.10% gene-space coverage, supporting projection feasibility rather than source-localization accuracy.

Supplementary Tables

Table S1. Internal validation design

Validation	Data	Held-out unit	Route tested	Reported endpoints	Denominator policy
Network LOSO	Bo2023 macaque brain RNA-seq	Single sample	Projected VSD vs logCPM/native VSD Network scoring	Network Top1, Network Top3, median true-rank	All 819 samples
Network LOMO	Bo2023 macaque brain RNA-seq	One monkey	Projected VSD vs logCPM/native VSD Network scoring	Network Top1, Network Top3, median true-rank	All 819 samples
Formal three-tier LOSO	Bo2023 macaque brain RNA-seq	Single sample	Projected-VSD Network beam plus logCPM group/exact rerank	Network, resolution-group and exact-region Top1/Top3	Network n=819; group/exact n=814
Formal three-tier LOMO	Bo2023 macaque brain RNA-seq	One monkey	Fold-local Network beam plus logCPM group/exact rerank	Network, resolution-group and exact-region Top1/Top3	Network n=819; group/exact n=812

Table S2. Internal validation results

Dataset	Route	Endpoint	Evaluated n	Top1	Top3	Median true-rank	Interpretation
Bo2023 LOSO	Projected VSD Network	Network	819	58.00%	91.58%	1.0	Supports projected-VSD Network beam
Bo2023 LOMO	Projected VSD Network	Network	819	53.72%	91.33%	1.0	Supports donor-level Network beam
Bo2023 LOSO	Formal hybrid	Network	819	58.24%	92.19%	1.0	Primary endpoint across all Network-evaluable samples
Bo2023 LOSO	Formal hybrid	Resolution group	814	44.47%	72.36%	2.0	Main region-level endpoint among reference-supported samples
Bo2023 LOSO	Formal hybrid	Exact region	814	22.48%	45.33%	4.0	Exploratory ranking among reference-supported samples
Bo2023 LOMO	Formal hybrid	Network	819	57.75%	91.21%	1.0	Cross-monkey support
Bo2023 LOMO	Formal hybrid	Resolution group	812	41.38%	69.09%	2.0	Main region-level endpoint among reference-supported samples
Bo2023 LOMO	Formal hybrid	Exact region	812	22.17%	42.36%	5.0	Exploratory ranking among reference-supported samples

Table S3. External validation design

Dataset	Sample type	n	Truth label type	Allowed conclusion
AHBA	Human normal brain RNA-seq	242 total; 233 supported; 91 exact-evaluable	Mapped anatomical labels	Cross-species mapped-label validation
TCGA/BraTS	Glioma tissue RNA-seq with MRI-derived labels	65 patients	Human imaging labels	Coarse anatomical consistency only
GSE189919	External count matrix	51 samples	No patient-level anatomical truth	Projection feasibility only

Table S4. External validation results

Dataset	Route	Endpoint	Top1	Top3	Conclusion
AHBA	Hybrid	Network	74.68%	94.42%	Strong mapped-label Network support
AHBA	Hybrid	Resolution group	36.26%	67.03%	Moderate group-level support
AHBA	Hybrid	Exact region	24.18%	42.86%	Only exact-evaluable mapped labels
AHBA	logCPM baseline	Exact region	17.58%	30.77%	Below hybrid exact Top3
AHBA	Projected VSD only	Exact region	10.99%	29.67%	Below hybrid exact Top3
TCGA/BraTS	Hybrid	Network	15.38%	40.00%	Coarse consistency only
TCGA/BraTS	Hybrid	Broad anatomy	13.85%	64.62%	Coarse consistency only
GSE189919	Projection feasibility	Projector gene overlap	15622/21668	72.10%	No accuracy claim

Table S5. Figure and table index

Item	Content	Use in manuscript
Figure 1	Formal route diagram and validation summary; numeric source in Figure1_validation_summary.csv	Main Application Note figure
Table S1	Internal validation design	Documents LOSO/LOMO operations
Table S2	Internal validation results	Reports Network, group and exact-region metrics
Table S3	External validation design	Documents label support and allowed conclusions
Table S4	External validation results	Reports AHBA, TCGA/BraTS and GSE189919 outcomes
Table S5	Figure/table index	Indexes manuscript figure and supplementary tables
Table S6	Claim boundaries	Defines unsupported interpretations and permitted claims

Table S6. Claim boundaries

Avoid	Use
Projected VSD creates a new Bo2023 atlas.	Only query profiles are projected for Network beam generation.
Projected VSD is best for exact-region inference.	Projected VSD supports Network beam; logCPM supports downstream reranking.
TCGA/BraTS validates Bo2023 exact regions.	TCGA/BraTS supports coarse anatomical consistency.
GSE189919 validates accuracy.	GSE189919 verifies projection feasibility only.
Exact Top1 is deterministic localization.	Exact outputs are exploratory local candidate rankings.